

PLEASE REFER TO ENGINEERING PACKET
FOR ALL STRUCTURAL DETAILS

1. GENERAL CONTRACTOR TO CONFORM TO ALL LOCAL AND STATE BUILDING CODE REQUIREMENTS.
2. GENERAL CONTRACTOR TO VERIFY ALL CONDITIONS AND DIMENSIONS SHOWN ON THE DRAWINGS AND NOTIFY THE ENGINEER OF ANY DISCREPANCIES.
3. THE ENGINEER IS RESPONSIBLE ONLY FOR INFORMATION SHOWN ON THE CERTIFIED ENGINEER'S DRAWINGS. THE DESIGN AND LAYOUT OF ALL OTHER INFORMATION IS THE RESPONSIBILITY OF OTHERS AND MUST CONFORM TO THE MASSACHUSETTS BUILDING CODE REQUIREMENTS. REFER TO STRUCTURAL ENGINEERING BY OTHERS FOR CERTIFIED BEAM CALCULATIONS AND CERTIFIED WIND DESIGN DETAILS.
4. ALL HEATING, PIPING, INSULATION, ELECTRICAL, FIREPROOFING AND OTHER REQUIREMENTS ARE THE RESPONSIBILITIES OF OTHERS.
5. NOTIFY THE ENGINEER OF ANY ARCHITECTURAL MODIFICATIONS OR DIMENSION CHANGES THAT MAY AFFECT THE STRUCTURAL DESIGN.

3. ALL STEEL BEAMS SHALL BE NEW STEEL CONFORMING TO THE A.I.S.C. SPECIFICATIONS FOR DESIGN, FABRICATION AND ERECTION OF STRUCTURAL STEEL FOR BUILDINGS AND A.S.T.M. 9 SPECIFICATION A53, TYPE "E" OR "S", GRADE "B", WITH A MINIMUM YIELD STRESS OF 35 K.S.I.
4. ALL SHOP AND FIELD WELDS SHOWN SHALL BE MADE BY APPROVED CERTIFIED WELDERS AND SHALL CONFORM TO THE A.W.S. CODE FOR BUILDINGS. ALL WELDS SHALL DEVELOP THE FULL STRENGTH OF THE MATERIAL BEING WELDED. USE EXX 16 ELECTRODES.
5. NO PERMANENT CONNECTIONS SHOULD BE MADE UP UNTIL THE STRUCTURE HAS BEEN PROPERLY ALIGNED. PROVIDE TEMPORARY BRACING AS REQUIRED.
6. STEEL FABRICATOR IS RESPONSIBLE FOR FINAL LENGTHS, CONNECTION DETAILS AND DESIGN IN ACCORDANCE WITH THE MINIMUM REQUIREMENTS OF THE LATEST EDITION OF THE A.I.S.C. DETAILING MANUAL. SUBMIT SHOP DRAWINGS WITH ALL DETAILS TO THE GENERAL CONTRACTOR PRIOR TO FABRICATION.
7. USE 1/2" MINIMUM CAP PLATE AND BASE PLATES (6X6 MINIMUM) FULLY WELDED ALL AROUND AT COLUMNS WITH 3/16" FILLET WELD, OR AS OTHERWISE SPECIFIED ON THE DRAWINGS. ALL STEEL COLUMN EXTERIOR BASE PLATE SHALL BE BOLTED TO THE CONCRETE FOUNDATIONS WITH 4-5/8" DIAMETER ANCHOR BOLTS.
8. ALL STEEL SHALL HAVE TWO COATS OF RUST-INHIBITOR PRIMER PAINT. TOUCH UP ALL WELDS, SCRATCHES OR SCRAPES IN PAINT AFTER ERECTION.
9. STEEL BEAM MAY BE SPICED AT STEEL COLUMN CAP PLATE WITH A MAXIMUM GAP BETWEEN BEAMS OF 1/4". USE 1/4" TIE PLATE WELDED TO WEBS.
10. FRAME JOISTS TO TOP OF BEAM ON A 2X8 TOP NAILER THRU-BOLTED WITH 1/2" DIAMETER BOLTS STAGGERED AT 24" O.C. JOISTS TO BE ANCHORED TO THE TOP NAILER WITH SPM50N H4 HURRICANE CLIPS. FLUSH FRAME JOISTS TO THE FULL DEPTH WEB BLOCKING FASTENED TO THE BEAM WITH 1/2" DIAMETER THRU-BOLTS AT 24" O.C. STAGGERED TOP AND BOTTOM.

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1. ALL FRAMING LUMBER SHALL BE S.P.F. (SPURCE-PINE-FIR) GRADE N/12 OR APPROVED EQUAL (UNLESS OTHERWISE SPECIFIED) AND SHALL MEET THE REQUIREMENTS OF THE AMERICAN FOREST AND PAPER ASSOCIATION. THE MINIMUM ALLOWABLE BENDING STRESS (FB) SHALL BE 875 P.S.I. THE MINIMUM ALLOWABLE COMPRESSION STRESS (FC) SHALL BE 425 P.S.I. THE MINIMUM ALLOWABLE MODULUS OF ELASTICITY (E) SHALL BE 1,400,000 P.S.I. OTHER FRAMING MATERIAL FOR INTERIOR NON-LOAD BEARING STUDS MAY BE SUBSTITUTED ONLY UPON APPROVAL OF THE ENGINEER.
2. ALL PRESSURE TREATED (CCA TREATED) DIMENSIONAL FRAMING LUMBER SHALL BE SOUTHERN YELLOW PINE GRADE NO. 2. THE MINIMUM ALLOWABLE BENDING STRESS (FB) SHALL BE 1,050 P.S.I. THE MINIMUM ALLOWABLE COMPRESSION STRESS (FC) SHALL BE 565 P.S.I. THE MINIMUM ALLOWABLE MODULUS OF ELASTICITY (E) SHALL BE 1,600,000 P.S.I.
3. ALL LVLS TO BE MANUFACTURED BY TRUS JOIST, GEORGIA PACIFIC OR APPROVED EQUAL. THE MINIMUM ALLOWABLE BENDING STRESS (FB) SHALL BE 2,900 P.S.I. THE MINIMUM ALLOWABLE COMPRESSION STRESS (FC) PERPENDICULAR TO THE GRAIN SHALL BE 750 P.S.I. THE MINIMUM ALLOWABLE MODULUS OF ELASTICITY (E) SHALL BE 2,000,000 P.S.I. ALL PARALAMS EXPOSED TO THE WEATHER SHALL BE PRESSURE TREATED (CCA TREATED). INSTALL MICROLAMS AND PARALAMS IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS.
4. PARALAM (LAM) POSTS SHALL HAVE AN ALLOWABLE COMPRESSION STRESS OF 2900 PSI AND A MODULUS OF ELASTICITY OF 2,000,000.
5. USE 3/4" TONGUE AND GROOVE STRUCTURAL GRADE F1 PLYWOOD FLOOR SHEATHING, 5/8" EXTERIOR STRUCTURAL GRADE F1R (C.D.X.) PLYWOOD ROOF SHEATHING AND 1/2" EXTERIOR STRUCTURAL GRADE F1R (C.D.X.) AT WALLS. ALL JOINTS SHALL BE BLOCKED WITH LUMBER OR OTHER APPROVED SUPPORTS.
6. ALL EXTERIOR AND INTERIOR STUD WALLS TO BE 2X4 MINIMUM @ 16" O.C. UNLESS NOTED OTHERWISE.
7. PROVIDE ADEQUATE WALL RESISTANCE TO RAKING BY DIAGONAL CORNER WIND BRACING ANCHORED TO SILL PLATES.
8. PROVIDE SOLID BLOCKING BETWEEN FLOOR JOISTS AND/OR DOUBLE ALL JOISTS UNDER EACH PARTITION.
9. USE FULLY NAILED METAL CONNECTORS (TECO, SIMPSON OR EQUAL), JOIST OR BEAM HANGERS WHEN JOISTS OR BEAMS FRAME INTO OTHER JOISTS OR BEAMS. PROVIDE METAL POST CAPS AND BASES FOR ALL POSTS.
10. FOR NONBEARING ROUGH WINDOW OPENINGS AND INTERIOR DOOR OPENINGS UP TO 3 FEET, USE 2-2X6 HEADER BEAMS. FROM 3 FEET TO 5 FEET, USE 2-2X8 HEADER BEAMS AND FROM 5 FEET TO 1 FEET, USE 2-2X10 HEADER BEAMS AND USE LVLS FOR SPANS EXCEEDING 1 FEET, EXCEPT AS NOTED OTHERWISE ON THE PLANS OR SPECIFICATIONS. USE TRIPLES FOR 2X6 WALLS. IF LVLS ARE SPECIFIED ON THE PLANS, PROVIDE DOUBLE JACK STUD SUPPORTS OR AS OTHERWISE SPECIFIED ON THE PLAN.
11. ALL FRAMING TO BE INSTALLED IN ACCORDANCE WITH THE MASSACHUSETTS BUILDING CODE REQUIREMENTS AND GENERAL FRAMING PRACTICE AS DETAILED IN THE "ARCHITECTURAL GRAPHICS STANDARDS", BY RAMSEY & SLEEPER.
12. ALL PLYWOOD FLOOR SHEATHING SHALL BE GLUED TO SUPPORTING WOOD FRAMING MEMBERS USING AMERICAN PLYWOOD ASSOCIATION (A.P.A.) GLUED FLOOR SYSTEM. WOOD GLUE TO BE CONCRETE ENCL. PL400 SUBFLOOR CONSTRUCTION ADHESIVE, OR APPROVED EQUAL.
13. ALL WALL STUDS TO ALIGN WITH FLOOR JOISTS AND ROOF RAFTERS.
14. THE CROSS WALLS AND THE BEAMS ARE TO PROVIDE THE LATERAL RESTRAINT FOR THE BUILDINGS AND SHOULD BE SECURELY ATTACHED AT EACH END AND/OR TO THE EXTERIOR WALLS.
15. BUILT-UP BEAMS (3 PIECES MAXIMUM) USING CONVENTIONAL FRAMING LUMBER SHALL BE FULLY SPIKED TOGETHER WITH 2-10D NAILS AT 8" O.C. AND LVLS WITH 2-16D NAILS (TOP AND BOTTOM) AT 8" O.C., OR AS OTHERWISE NOTED ON THE DRAWINGS, OR AS RECOMMENDED BY THE MANUFACTURER.
16. ALL NAILS, FASTENERS AND CONCRETE EXPOSED TO THE WEATHER SHALL BE HOT-DIP GALVANIZED.
17. ALL LUMBER THAT COMES IN CONTACT WITH CONCRETE SHALL BE PRESSURE TREATED.

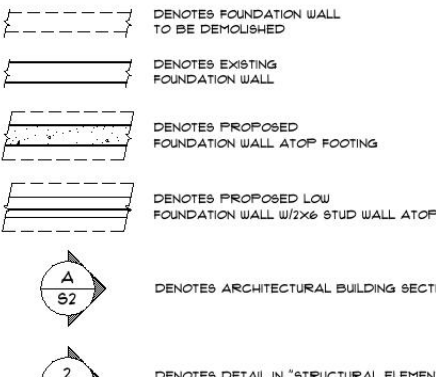
9. SPREAD FOOTINGS SHALL BEAR LEVEL ON UNDISTURBED SOIL HAVING AN ALLOWABLE BEARING CAPACITY OF TWO TONS PER SQUARE FOOT.
10. IF BEARING MATERIALS WITH A LOWER BEARING CAPACITY THAN TWO TONS PER SQUARE FOOT ARE ENCOUNTERED AT THE SPECIFIED ELEVATIONS, THE UNDERLYING UNSUITABLE MATERIAL SHALL BE REMOVED AND REPLACED WITH SUITABLE MATERIAL TO BE APPROVED BY THE ENGINEER/ARCHITECT.
11. THE ENGINEER ASSUMES NO RESPONSIBILITY FOR THE VALIDITY OF SUBSURFACE CONDITIONS.
12. NO FOUNDATION SHALL BE PLACED IN WATER OR ON FROZEN GROUND.
13. FOOTINGS SHALL BE PROTECTED AGAINST FROST UNTIL PROJECT IS COMPLETED.
14. BACKFILL UNDER ANY PORTION OF THE FOOTINGS AND SLABS SHALL BE COMPACTED IN 6" LIFTS OF 95% COMPACTED GRAVEL AS APPROVED BY THE ENGINEER.
15. CONCRETE WORK SHALL CONFORM TO THE LATEST AMERICAN CONCRETE INSTITUTE CODE FOR "BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE" AND "SPECIFICATIONS FOR STRUCTURAL CONCRETE FOR BUILDINGS".
16. CONCRETE FOUNDATION WALLS AND FOOTINGS SHALL HAVE A MINIMUM COMPRESSIVE STRENGTH OF 3,000 P.S.I. AT 28 DAYS AND 3,500 P.S.I. FOR SLABS, WITH A SLUMP OF NO MORE THAN 4" AND AIR ENTRAINMENT OF 4-6%. THE USE OF CALCIUM CHLORIDE IS NOT PERMITTED. PROVIDE PROPER CONCRETE PROTECTION FOR HEAT IN COLD WEATHER AND MAINTAIN PROPER CURING PROCEDURES IN ACCORDANCE WITH THE A.C.I.
17. STEEL REINFORCEMENT SHALL CONFORM TO A.S.T.M. 615, GRADE 60.
18. ALL CONCRETE SLABS ON THE GROUND SHALL BE REINFORCED WITH 6x6-10/10 (MIN.) WELDED WIRE FABRIC PLACED AT MID-DEPTH, OR AS OTHERWISE SHOWN ON THE DRAWINGS. WELDED WIRE FABRIC REINFORCEMENT SHALL CONFORM TO A.S.T.M. A185, AND SHALL LAP 6" MINIMUM OR ONE SPACE, WHICHEVER IS LARGER, AND SHALL BE WIRED TOGETHER. PROVIDE SUFFICIENT CHAIR OR SUPPORT BARS AS NECESSARY TO POSITION WELDED WIRE FABRIC.
19. WHERE CONTINUOUS BARS ARE CALLED FOR THEY SHALL BE RUN CONTINUOUSLY AROUND CORNERS AND LAPPED AT NECESSARY SPICES OR HOOKED AT DISCONTINUOUS ENDS. LAPS SHALL BE 40 BAR DIAMETERS, UNLESS OTHERWISE SHOWN.
20. NOTIFY BUILDING DEPARTMENT FOR INSPECTION OF COMPLETED INSTALLATION OF REINFORCEMENT AT LEAST 24 HOURS PRIOR TO SCHEDULED PLACEMENT OF CONCRETE.
21. PLACEMENT OF CONCRETE POURS FOR FOUNDATION WALLS SHOULD HAVE A VERTICAL 2"x4" KEY WITH CONTINUOUS REINFORCING (40 BAR DIAMETER MINIMUM) THRU THE CONSTRUCTION JOINT.
22. ALL REINFORCING BARS SHALL BE COLD BENT IN ACCORDANCE TO THE PROPER RADII ESTABLISHED BY THE AMERICAN CONCRETE INSTITUTE. UNDER NO CONDITIONS SHALL HEAT BE APPLIED TO THE BARS TO OBTAIN BENDS.
23. THE USE OF CONTROL JOINTS IN THE SLAB IS RECOMMENDED TO CONTROL CRACKING. SAW CUT TO A DEPTH ONE HALF INCH NOT-TO-EXCEED 10 FEET BY 10 FEET.
24. DAMP PROOF ALL FOUNDATION WALLS BELOW GRADE, OTHER THAN FROST WALLS.

WALL TO BE DEMOLISHED	
EXISTING STUD WALL	
PROPOSED STUD WALL	
OBJECT ABOVE	
OBJECT BELOW	

FIRST FLOOR LIVING AREA	=	FT ²
SECOND FLOOR LIVING AREA	=	FT ²
UNFINISHED ATTIC FLOOR AREA	=	FT ²
TOTAL FINISHED LIVING AREA:	=	FT ²

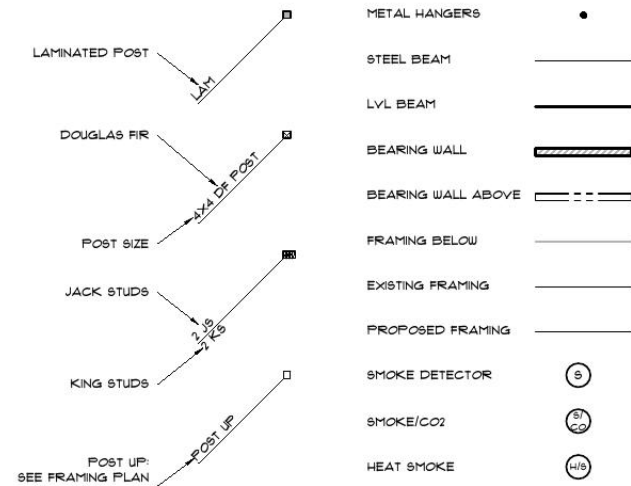
A0	COVER PAGE
A1	NOTES AND LEGENDS
A2	FRONT ELEVATION
A2.1	REAR ELEVATION
A2.2	RIGHT SIDE ELEVATION
A2.3	LEFT SIDE ELEVATION
A3	FIRST FLOOR PLAN
A4	SECOND FLOOR PLAN
A5	ATTIC FLOOR PLAN
A6	ROOF PLAN
A7	BUILDING SECTION "A-A"
A8	BUILDING SECTIONS "B-B" & "D-D"
A9	BUILDING SECTION "C-C"
A10	CUPOLA DETAIL
B1	FOUNDATION PLAN

- 1) SEE "STRUCTURAL ENGINEERING AND WFCM ANALYSIS" BOOKLET NO 19- BY _____ ENGINEERING, LLC DATED AUG/2019 FOR ADDITIONAL NOTES AND DETAILS
- 2) ALL NEW FOUNDATION WALLS SHALL BE DAMP PROOFED WITH A BITUMINOUS COATING.
- 3) 6 SQ. FT. OF VENTILATION REQUIRED FOR EVERY 1,500 SQ. FT. OF BASEMENT AREA.
- 4) OPENING FOR UNDER-FLOOR VENTILATION: 1 SQ. FT. OPENING/150 SQ. FT. OF CRAWL SPACE AREA



1) SEE "STRUCTURAL ENGINEERING AND WFCM ANALYSIS"
BOOKLET NO. 19-___ BY _____ ENGINEERING, LLC
DATED AUG/___/2019 FOR ADDITIONAL NOTES AND DETAILS

2) ALL FRAMING MEMBERS SHALL BE FASTENED IN ACCORDANCE WITH MANUFACTURER'S
RECOMMENDATIONS AND THE COMMONWEALTH OF MASSACHUSETTS 106 CMR



2024

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